

# **THYROID DISORDERS**

Key Family Physicians: Dr Cheah Ming Hann / Dr Chew Eng Lee

Specialist Advisor: Dr Khoo Chin Meng (Senior Consultant, Division of Medicine, National University Hospital)

*Acknowledgement for their inputs: NUHSP: Ms Neo Ying Fang*

*Reviewed and updated in December 2023 by Dr Cheah Ming Hann*

# CONTENTS

## HYPOTHYROIDISM

|                                                                |          |
|----------------------------------------------------------------|----------|
| <b>Introduction .....</b>                                      | <b>3</b> |
| Common Causes.....                                             | 3        |
| Hashimoto's Thyroiditis (Chronic Autoimmune Thyroiditis) ..... | 3        |
| <b>Screening and Diagnosis .....</b>                           | <b>3</b> |
| Symptoms and Signs .....                                       | 4        |
| <b>Management and Follow-up .....</b>                          | <b>4</b> |
| Initial Management of Hypothyroidism .....                     | 4        |
| Goals of Therapy .....                                         | 5        |
| Pharmacological Treatment of Hypothyroidism .....              | 6        |
| <b>Special Situations .....</b>                                | <b>8</b> |
| Subclinical Hypothyroidism .....                               | 8        |
| Post-Thyroidectomy Hypothyroidism .....                        | 8        |
| Urgent and Emergency Situations .....                          | 8        |
| <b>Referrals.....</b>                                          | <b>9</b> |

## HYPERTHYROIDISM

|                                                                        |           |
|------------------------------------------------------------------------|-----------|
| <b>Introduction .....</b>                                              | <b>10</b> |
| Causes .....                                                           | 10        |
| <b>Screening and Diagnosis .....</b>                                   | <b>11</b> |
| Symptoms and Signs .....                                               | 11        |
| <b>Management and Follow-up .....</b>                                  | <b>12</b> |
| Initial Management of Hyperthyroidism depends on the TFT results. .... | 12        |
| Pharmacological Treatment of Thyrotoxicosis .....                      | 13        |
| Table 2: Modalities of Treatment for Thyrotoxicosis .....              | 15        |
| Table 3: Medical Treatment for Thyrotoxicosis .....                    | 16        |
| <b>Special Situations .....</b>                                        | <b>17</b> |
| Subclinical Hyperthyroidism .....                                      | 17        |
| Detection and Treatment of Complications.....                          | 17        |
| Urgent and Emergency Situations .....                                  | 17        |
| <b>Referrals.....</b>                                                  | <b>18</b> |
| <b>Abbreviations .....</b>                                             | <b>19</b> |
| <b>References .....</b>                                                | <b>19</b> |

## **HYPOTHYROIDISM**

### **INTRODUCTION**

In community surveys, the prevalence of overt hypothyroidism varies from 0.1 to 2%. The prevalence of subclinical hypothyroidism is higher, from 4 to 10% of adults, with possibly higher frequency in older women. Hypothyroidism is 5 to 8 times more common in women than men.

### **Common Causes**

1. Post radioiodine
2. Post-thyroidectomy
3. Over-treated thyrotoxicosis cases.
4. Thyroiditis - Hashimoto's disease (31 to 60% of hypothyroidism cases), subacute thyroiditis, post-partum thyroiditis
5. Drugs, e.g., lithium, amiodarone
6. Central hypothyroidism due to external irradiation or hypothalamic / pituitary disorder.

Primary thyroid disease accounts for 95% of cases of hypothyroidism. It is important to exclude secondary hypothyroidism which requires referral to the endocrinologist.

Identifying the causes is important as hypothyroidism may be transient (e.g., subacute and post-partum thyroiditis) and require short-term or no treatment.

### **Hashimoto's Thyroiditis (Chronic Autoimmune Thyroiditis)**

- Bilateral goitre – firm, rubbery
- Positive thyroid peroxidase antibodies (anti-TPO) and/or thyroglobulin antibodies (anti-TG)
- May present as postnatal depression.
- May be associated autoimmune conditions – diabetes mellitus, pernicious anaemia, vitiligo, Addison's disease.

### **SCREENING AND DIAGNOSIS**

The clinical manifestations of hypothyroidism are highly variable, depending on the age at onset and the duration and severity of thyroid hormone deficiency. Due to the lack of specificity of the typical clinical manifestations, the diagnosis of hypothyroidism relies on laboratory testing, i.e., thyroid function test (free T4 and TSH).

## Symptoms and Signs

| Symptoms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Physical Findings                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Lethargy</li><li>• Cold intolerance</li><li>• Symptoms of depression</li><li>• Memory and mental impairment, poor attention span</li><li>• Hoarse voice, dry skin, puffy face, hair loss</li><li>• Non-specific aches and pains</li><li>• Unexplained weight gain</li><li>• Constipation</li><li>• Paraesthesia, nerve entrapment syndromes</li><li>• Menorrhagia</li><li>• Erectile dysfunction / subfertility</li><li>• Pubertal delay / growth failure in the young</li><li>• Asymptomatic</li></ul> | <ul style="list-style-type: none"><li>○ Obesity / overweight</li><li>○ Dry, cool skin</li><li>○ Coarse, dry and brittle hair</li><li>○ Goitre</li><li>○ Sinus bradycardia</li><li>○ Umbilical hernia in the young</li><li>○ Non-pitting oedema of hands, face or ankles, periorbital swelling</li><li>○ Delayed deep tendon reflexes<br/>~ classically of the ankle reflex</li></ul> |

## MANAGEMENT AND FOLLOW-UP

### Initial Management of Hypothyroidism

1. Thyroid function test (TFT) to ascertain thyroid status: **Thyroid-Stimulating Hormone (TSH) assay is the most sensitive test for diagnosis.**
2. Presence of high titres of **anti-TPO** is suggestive of Hashimoto's thyroiditis.  
Titres may decrease over time in long-standing Hashimoto's thyroiditis.
3. Investigation for complications:
  - Full blood count for anaemia (where relevant, also to look out for B12 deficiency)
  - Lipid panel for high cholesterol
  - Electrolytes for hyponatraemia (where relevant)

### Thyroid Function Test (TFT)

|            |                                                                                                                                                                           |                          |                                                                                                                                                                                                                      |                            |                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ft4</b> | ↓                                                                                                                                                                         | ↓                        | ↓                                                                                                                                                                                                                    | N                          | N                                                                                                                                                                           |
| <b>TSH</b> | ↑                                                                                                                                                                         | ↓                        | N                                                                                                                                                                                                                    | ↑                          | N                                                                                                                                                                           |
|            | Hypothyroidism                                                                                                                                                            |                          |                                                                                                                                                                                                                      | Subclinical Hypothyroidism | Functionally Normal                                                                                                                                                         |
|            | Primary Hypothyroidism                                                                                                                                                    | Secondary Hypothyroidism |                                                                                                                                                                                                                      |                            |                                                                                                                                                                             |
|            | <p>a) Check anti-TPO if suspicious of Hashimoto's disease (e.g., no past history of radioiodine treatment or thyroidectomy).</p> <p>b) Start L-thyroxine replacement.</p> |                          | <p>Any evidence of acromegaly, Cushing's disease, adrenal insufficiency, unexplained infertility or symptoms or signs of high prolactin. Any ophthalmoplegia. Refer Endocrinologist. Do not start L-thyroxine.**</p> |                            | <p>Prevalence is higher with increasing age.</p> <p>a) Repeat ft4, TSH in 3 to 6 months.</p> <p>b) Individualised treatment (whether to start L-thyroxine replacement).</p> |
|            |                                                                                                                                                                           |                          |                                                                                                                                                                                                                      |                            | <p>Not hypothyroidism</p> <p>Consider other diagnosis.</p>                                                                                                                  |

**\*\*** In secondary hypothyroidism, starting L-thyroxine when there is concomitant ACTH deficiency (in pituitary disorders) can precipitate an Addisonian crisis.

### Goals of Therapy

1. Ameliorate symptoms
2. Normalise TSH secretion
3. Reduce goitre size (if present)
4. Avoid overtreatment (iatrogenic thyrotoxicosis)

Aim to keep serum TSH within the normal reference range.

## **Pharmacological Treatment of Hypothyroidism**

### **Replacement with L-thyroxine in Primary Hypothyroidism**

- The starting dose is dependent on age, body weight, history of coronary heart disease and severity and duration of the hypothyroidism. Average replacement dosage is 1.6 mcg/kg lean body weight. Some patients may require higher dose.
- Start at low dose and **titrate gradually based on symptoms and free T4 (by increasing 25 mcg every 3 to 4 weeks)** if the patients are elderly (e.g., > 60 years), have coronary heart disease or have very mild hypothyroidism (i.e. mildly low fT4 and mildly raised TSH).
- For young patients or patients without history of coronary heart disease, full replacement dose can be given without stepwise process.
- Pregnant women may need adjustment to higher doses for L-thyroxine replacement. They should be monitored monthly. They should be referred to endocrinologists for further treatment, where needed.

### **Caution:**

- Take L-thyroxine on empty stomach with plain water, 60 minutes before breakfast. Advise patients to take supplements such as calcium or iron or soya products before lunch.
- If patient gets angina on increasing L-thyroxine dose, a cardiology referral should be made to investigate for underlying ischaemic heart disease.

### Starting dose of L-thyroxine:

12.5 mcg - 25 mcg OM for elderly patients or patients with coronary heart disease

25 mcg - 50 mcg OM for adult patients or patients without coronary heart disease



Repeat TFT in 6 - 8 weeks.

Reassess symptoms after starting L-thyroxine therapy. Symptoms may abate after 2-3 weeks but steady state TSH concentrations are not achieved for at least 6 weeks.

Serum TSH is the most reliable endpoint for primary hypothyroidism treatment while free T4 (fT4) should guide therapy in central hypothyroidism.



| TSH | High                                                                                                                                                                                                                                                                                                                                                 | Low                                                                                                                                                                                                      | Normal                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <ul style="list-style-type: none"><li>• Check compliance with L-thyroxine.</li><li>• Exclude drug interaction with sucralfate, calcium &amp; iron supplements, proton pump inhibitors.</li><li>• If compliant with medication and sub-optimal treatment, increase dose by 12.5-25 mcg OM each visit.</li><li>• Repeat TFT in 6 to 8 weeks.</li></ul> | <ul style="list-style-type: none"><li>• Over-replacement.</li><li>• Decrease dose – may need to give alternating doses e.g., L-thyroxine 50, 75 mcg EOD.</li><li>• Repeat TFT in 6 to 8 weeks.</li></ul> | <ul style="list-style-type: none"><li>• Maintenance dose achieved.</li><li>• fT4 levels may be high normal or slightly &gt; normal.</li><li>• Follow up in 6 months, then yearly TFT.</li><li>• Stable patients with no co-morbidities can be monitored with yearly TFT if 2 previous TFTs done at least 6 months apart show normal fT4 and normal TSH. They can be advised to return earlier for review if any hypo or hyperthyroid symptoms.</li></ul> |

## **SPECIAL SITUATIONS**

### **Subclinical Hypothyroidism**

Diagnosis is based on biochemical testing with a normal free T4 and an elevated TSH.

Progression to overt hypothyroidism varies from 3 to 20%, the risks being higher in older individuals, those with higher antibody titres, and previous radioiodine therapy.

Treatment is recommended for those patients whose TSH is  $> 10$  mU/L.

Treatment is individualized (observation or treatment) for those patients whose TSH is between 4.5 to 10 mU/L.

Treatment is generally given in the presence of positive anti-TPO, goitre, a strong family history of autoimmune thyroid disease or infertility.

### **Post-Thyroidectomy Hypothyroidism**

Look for reason for thyroidectomy.

In Thyroid Cancer, consider referral back to surgeon / endocrinologist. NUP does not perform serum thyroglobulin levels which is recommended at intervals to detect recurrence.

If patients decline to be referred back, maintain thyroid suppression therapy to achieve TSH target levels as per initial stepdown memo.

In the absence of previous memos,

- For metastatic / invasive cancers, aim TSH  $< 0.01$  mU/L.
- For other cancers, aim TSH 0.05 to 0.5 mU/L.

### **Urgent and Emergency Situations**

**Myxoedema coma** is an endocrine emergency that needs urgent referral to the Emergency Department. It can occur due to severe long-standing hypothyroidism or precipitated by an acute event in poorly controlled hypothyroid patient such as infection, myocardial infarction, cold exposure, surgery, or administration of sedative drugs especially opioids.

## **REFERRALS**

Consult the relevant specialist in the following situations:

### **Endocrinology**

- Children and infants (hypothyroidism may be transient in children.)
- Patients in whom it is difficult to render and maintain a euthyroid state
- Pregnant patients (Direct Access referral within 2 weeks would be indicated)
- Suspicion of secondary hypothyroidism

### **Cardiology**

- Patients with cardiac disease

### **Endocrine Surgery**

- Presence of enlarging nodule

Consider referral to the Emergency Department in severely hypothyroid patients who are symptomatic with suspicion of impending myxoedema coma (e.g., significant bradycardia, signs and symptoms of cardiac failure).

## **THYROTOXICOSIS / HYPERTHYROIDISM**

### **INTRODUCTION**

Hyperthyroidism is more common in women than men (5:1 ratio). The overall prevalence is approximately 1.3% and increases to 4 to 5% in older women. It is also more common in smokers. Graves' disease is seen most often in younger patients while toxic nodular goitre is more common in older population.

It is essential that the correct cause be identified as appropriate therapy depends upon the underlying mechanism of the hyperthyroidism.

### **Causes**

#### **Primary (Thyroid)**

- Graves' disease is the commonest cause of hyperthyroidism
- Toxic multinodular goitre
- Toxic adenoma
- Painful subacute thyroiditis (de Quervain's)
- Post-partum thyroiditis
- Hashimoto's thyroiditis (initial phase)

#### **Secondary**

- Drugs (Iodine, Amiodarone, Lithium, excessive thyroxine replacement)
- Pituitary tumour secreting TSH or trophoblastic disease (rare)

**Table 1: Difference between Toxic Nodular Goitre and Graves' Disease**  
(\*Some patients may present with both Graves' disease and Toxic Nodular Goitre)

| <b>Toxic Nodular Goitre</b>                        | <b>Graves' Disease</b>                               |
|----------------------------------------------------|------------------------------------------------------|
| Usually older patients > 50 years                  | Usually younger patients < 40 years                  |
| Nodular enlargement of gland (asymmetrical)        | Diffuse goitre (symmetrical)                         |
| No Graves ophthalmopathy                           | Graves' ophthalmopathy                               |
| Association with autoimmune conditions is uncommon | Commonly associated with other autoimmune conditions |

## **SCREENING AND DIAGNOSIS**

A Thyroid Function Test (TFT) consisting of free T4 and TSH should be ordered when patients present with clinical signs and symptoms of hyperthyroidism.

## **Symptoms and Signs**

| Symptoms                                                                                                                                                                                                                                                                                                                                                           | Physical Findings                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Palpitations</li><li>• Nervousness and irritability</li><li>• Heat intolerance or increased sweating</li><li>• Weight loss despite good appetite</li><li>• Insomnia or mental disturbances</li><li>• Frequent bowel movements</li><li>• Exertional dyspnoea</li><li>• Fatigue</li><li>• Decreased menstrual flow</li></ul> | <ul style="list-style-type: none"><li>• Tremors</li><li>• Thyroid enlargement</li><li>• Eye signs: lid retraction, lid lag, Graves ophthalmopathy (periorbital edema, chemosis, ophthalmoplegia, exophthalmos)</li><li>• Atrial fibrillation (especially in the elderly) and tachycardia</li><li>• Dependent lower extremity oedema</li><li>• Proximal muscle weakness</li><li>• Brisk reflexes</li><li>• Alopecia</li><li>• Fragility fractures</li></ul> |

Older thyrotoxic patients may present in an atypical manner and may have predominant cardiovascular symptoms instead (apathetic thyrotoxicosis).

## **MANAGEMENT AND FOLLOW-UP**

**Initial Management of Hyperthyroidism** depends on the TFT results.

|                       |                        |                                           |                                                                      |                                    |                                                                                                                              |
|-----------------------|------------------------|-------------------------------------------|----------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Free T4</b>        | ↑                      | ↑                                         | N                                                                    | N                                  | ↑                                                                                                                            |
| <b>TSH</b>            | ↓                      | ↑                                         | ↓                                                                    | N                                  | N                                                                                                                            |
| <b>Interpretation</b> | Thyrotoxicosis         | Secondary thyrotoxicosis to be considered | Subclinical hyperthyroidism (no symptoms) or T3 toxicosis (symptoms) | Consider other causes for symptoms | Non-thyroidal illness, anti-psychotic drugs, anti-depressant, steroids, interference with assays, thyroid hormone resistance |
| <b>Action</b>         | Recall & treat patient | Refer Endocrinologist                     | Monitor 6 monthly and do TRAb; do fT3 if symptomatic                 | Reassess for other possible causes | Monitor, KIV refer Endocrinologist                                                                                           |

If patient is thyrotoxic (or if strong suspicion at initial presentation)

1. Give beta blockers (atenolol 25 – 50 mg OM or propranolol 20 mg BD) to alleviate tachycardia and hand tremors. Reduce heart rate to < 90/min if blood pressure allows. Beta blockers are contraindicated if patient has asthma or chronic obstructive lung disease.
2. Review patient early to review TFT in 1 to 2 weeks.
3. Start medical therapy once thyrotoxicosis is confirmed. Obtain baseline FBC and liver function test.
4. Explain to the patient the consequences of non-treatment, stopping treatment (symptoms and signs aggravation, cardiac effects including heart failure, irregular heart rhythm and stroke).
5. Check Thyroid Stimulating Hormone Receptor Antibodies (TRAb) to confirm Graves' disease or Thyroid Peroxidase antibodies (TPO Ab) if suspicious of Hashimoto's thyroiditis.

## **Pharmacological Treatment of Thyrotoxicosis**

Start **medical treatment** and follow protocol even if the patient eventually requires other modalities of treatment at specialist outpatient clinic (see [Table 2](#)).

### **Initiation of Treatment**

**It is important to educate patients that treatment with anti-thyroid drugs may take 18 to 24 months.**

1. **Compliance** is important to achieve remission.
2. Advise on possible adverse reactions of drugs.
  - Minor side effects include: abnormal sense of taste, pruritus, urticaria, fever, arthralgia. Serious complications are rare but include cholestatic jaundice, thrombocytopenia, lupus-like syndrome, hepatitis, agranulocytosis (incidence of few cases per thousand).
  - Advise patient to return immediately if he/she develops sudden high fever > 38° Celsius with sore throat, bruising, and arrange for a full blood count. Refer to hospital urgently if **agranulocytosis**.
  - Advise patient to stop the medication and return to see you if he / she develops rashes, bruising, jaundice, or joint pain.
3. Choice of Drugs:
  - Carbimazole or methimazole is the anti-thyroid drug of choice. It has a longer half-life, hence allows once-daily dosing.
  - Propylthiouracil is associated with a higher risk of liver dysfunction, thus it is recommended if your patient is not able to tolerate carbimazole or methimazole.
  - Propylthiouracil is preferred in women planning for pregnancy and during the first trimester of pregnancy.
  - Carbimazole or methimazole has been associated with aplasia cutis when used in early pregnancy although this is rare.
4. Aim is to achieve clinical and biochemical euthyroid status. It may take 3 to 6 months before euthyroid state is achieved.
5. Avoid fluctuating thyroid function test as it might aggravate thyroid eye disease.
6. Advise patient to stop smoking as it can aggravate thyroid eye disease.
7. Refer to ophthalmology if the patient has significant thyroid eye disease.

## Follow-Up Care

1. Follow-up schedule for medical treatment is as shown in [Table 3](#).
2. Monitor for symptoms resolution and TFT changes. The improvement in **TSH may lag behind the free T4 level** by a few months even though the patient is clinically euthyroid.
3. Some patients may have prolonged suppressed TSH, hence the dose of anti-thyroid drugs needs up-titration.
4. Patients can complain of weight gain, advise them that this is due to improved metabolism, and that they should watch the calorie intake.
5. Consideration to stop anti-thyroid drug if patient fulfils all these three criteria:
  - Normal thyroid size,
  - Negative TRAB level, **and**
  - Lowest drug dose before stopping, e.g., Carbimazole < 5mg daily.
6. Inform patient that the relapse rate is 50%. Upon cessation of anti-thyroid drugs, follow up with TFT at approximately 3 months and then 6 months. Thereafter, monitor TFT 1 year later. Relapse can occur at any time, hence important to advise patient to watch for symptoms of relapse and return earlier for TFT if thyrotoxic symptoms return.
7. Suspect Hashimoto's thyroiditis if free T4 decreases markedly to hypothyroid levels after initiation of treatment.

## Relapse

1. Restart medical treatment.
2. Consider radioactive iodine (RAI) therapy / surgery.
3. In patients who refuse RAI or surgery despite difficulty maintaining a normal TFT during the duration of treatment, long-term maintenance dose of anti-thyroid drugs may be necessary.
4. **Patients who have relapsed twice are at high risk of subsequent relapses and this should be accounted for in shared decision making if offering another trial off anti-thyroid medications.**

**Table 2: Modalities of Treatment for Thyrotoxicosis**

|                                    | Medical Treatment                                                                                                                                                                                                                                                                  | Radioactive-Iodine (RAI)                                                                                                                                                                                                                                                                                                                                                    | Surgical Treatment                                                                                                                                                                                                                                                                                                                      |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>When to Choose?</b>             | <ul style="list-style-type: none"><li>▪ Generally, first line therapy for Graves' disease.</li><li>▪ While preparing for surgery or radioactive iodine therapy.</li><li>▪ Pregnancy, elderly with comorbidities / surgical risk, previously operated or irradiated neck.</li></ul> | <ul style="list-style-type: none"><li>▪ Increasingly popular as first line therapy for Graves' disease.</li><li>▪ Failed medical or surgical therapy.</li><li>▪ Solitary toxic nodule.</li><li>▪ Patient's choice</li></ul>                                                                                                                                                 | <ul style="list-style-type: none"><li>▪ Huge goitre / Cosmesis</li><li>▪ Multinodular goitre and thyrotoxicosis</li><li>▪ Suspected malignancy</li><li>▪ Obstructive / compressive symptoms</li><li>▪ Patient's choice</li></ul>                                                                                                        |
| <b>When not to Choose?</b>         | <ul style="list-style-type: none"><li>▪ Adverse drug reactions</li><li>▪ Non-compliance by patient</li></ul>                                                                                                                                                                       | <ul style="list-style-type: none"><li>▪ During pregnancy</li><li>▪ Likely to become pregnant in the next 1 year.</li><li>▪ Contraindication in patients with active Graves' ophthalmopathy as RAI will worsen ophthalmopathy. Risk of worsening eye signs may be reduced with use of steroids.</li></ul>                                                                    | <ul style="list-style-type: none"><li>▪ If floridly toxic</li><li>▪ Scar is of major concern</li><li>▪ Previous thyroid surgery</li></ul>                                                                                                                                                                                               |
| <b>Advantages</b>                  | <ul style="list-style-type: none"><li>▪ Theoretically long-term remission without risk of hypothyroidism</li><li>▪ Avoids radiation exposure / surgical risks</li></ul>                                                                                                            | <ul style="list-style-type: none"><li>▪ Effective</li><li>▪ Simple treatment</li><li>▪ No anaesthesia risk or hospitalisation</li></ul>                                                                                                                                                                                                                                     | <ul style="list-style-type: none"><li>▪ Rapid restoration of euthyroidism</li><li>▪ Effective for large goitres</li></ul>                                                                                                                                                                                                               |
| <b>Disadvantages</b>               | <ul style="list-style-type: none"><li>▪ Slow effect (6 - 8 weeks)</li><li>▪ Prolonged treatment may be required</li><li>▪ Non-compliance</li><li>▪ Toxicity and side effects</li><li>▪ Frequent relapses</li></ul>                                                                 | <ul style="list-style-type: none"><li>▪ May require 2 to 3 months to take effect, during which anti-thyroid medication need to be continued but at lower dose.</li><li>▪ Occasionally patients with large goitres or severe hyperthyroidism may require more than one dose.</li><li>▪ Radiation thyroiditis (rare)</li><li>▪ Most patients develop hypothyroidism</li></ul> | <ul style="list-style-type: none"><li>▪ Operative morbidity / mortality</li><li>▪ Hospitalisation is required</li><li>▪ Neck scar / keloid formation</li><li>▪ Postoperative hypothyroidism, hypocalcaemia and recurrent nerve damage.</li><li>▪ Cost</li><li>▪ All patients develop hypothyroidism after total thyroidectomy</li></ul> |
| <b>Where to Refer if required?</b> | Endocrine SOC                                                                                                                                                                                                                                                                      | Nuclear medicine SOC (via Endocrine)                                                                                                                                                                                                                                                                                                                                        | Endocrine Surgery / ENT SOC                                                                                                                                                                                                                                                                                                             |

**Table 3: Medical Treatment for Thyrotoxicosis**

| Drug                                                                      | Strength per tab | Initiation of Treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Titration and Maintenance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | TFT and Follow-up Dates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Carbimazole</b>                                                        | 5 mg             | <p>20 - 60 mg/day<br/>(Carbimazole can be given once daily or in divided doses at higher total daily doses i.e., 15 mg BD for better efficacy.)</p> <p>Inform patient to observe for any symptoms suggestive of agranulocytosis (e.g., fever, sore throat) and return for review immediately if any are present.</p> <p><b>Suggested starting dose based on fT4 elevation above upper limit of normal (ULN)</b></p> <ul style="list-style-type: none"> <li>• 1 to 1.5 times: 5 to 15 mg / day</li> <li>• 1.5 to 2 times: 15 to 30 mg / day</li> <li>• 2 to 3 times: 30 to 60 mg / day</li> </ul> | <p>Decrease daily dose of carbimazole by 5 – 10 mg every visit according to clinical and biochemical response until euthyroid state.<br/><b>Consider larger decrease if fT4 declines sharply while on a moderate / high daily dose.</b></p> <p>Maintain carbimazole at 5 - 10 mg daily according to clinical and biochemical response.</p> <p>Generally, euthyroid state is achieved 3 to 6 months after start of initiation of treatment.</p> <p>May need maintenance dose of carbimazole or propylthiouracil if TSH persistently &lt; 1 mIU/L and high TRAb.</p> | <p><b>Second visit:</b><br/>Review clinically in 2 - 4 weeks.<br/>Perform full blood count if suspicious of agranulocytosis.<br/>Do TRAb if suspicious of Graves' disease and TPO Ab if suspicious of Hashimoto's thyroiditis.</p> <p><b>Third visit:</b><br/>Review in 6 to 8 weeks.<br/>Perform TFT one week before review.</p> <p>Patients with Hashimoto's thyroiditis may require low dose of propranolol for short periods of 2 to 3 months if toxic symptoms are present. The clinical outcome can be euthyroidism or hypothyroidism. Follow up every 6 - 8 weeks during titration and 3-monthly during maintenance. Do TFT one week before review. Titrate medication accordingly.</p> |
| <b>Propylthiouracil</b>                                                   | 50 mg            | <p>200 - 600 mg/day in divided doses every 8 hourly</p> <p>For women planning pregnancy and early pregnancy.</p> <p>Consider baseline Liver Function Test at initiation as propylthiouracil may cause transaminitis.</p>                                                                                                                                                                                                                                                                                                                                                                         | <p>Reduce propylthiouracil by 50 - 100 mg per day every 6 to 8 weeks according to clinical and biochemical response until euthyroid state. <b>Consider larger decrease if fT4 declines sharply while on a moderate / high daily dose.</b></p> <p>Maintain at 50 - 150 mg/day in divided doses every 8 - 12 hours.</p> <p>Routine LFTs are not required. However, any symptomatic patient (pruritic rash, jaundice, pale coloured stool) should have their LFTs performed.</p>                                                                                      | <p>Generally, most patients need to be on medication for 12 to 18 months. Patients with Graves' disease need to be treated for longer duration if TRAb is high towards end of 18 months.<br/>Consult senior doctor if patient relapses each time medication is tapered off at the end of 18 to 24 months.</p>                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Propranolol</b><br>(*contraindicated in patients with asthma and COPD) | 10, 40 mg        | 10 - 20 mg/dose every 8 hours (max 40 mg/dose)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Stop propranolol if adrenergic symptoms such as tremors or palpitations have resolved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## **SPECIAL SITUATIONS**

### **Subclinical Hyperthyroidism**

Less common than subclinical hypothyroidism

Clinical significance relates to 3 risk factors:

1. Progression to overt hyperthyroidism
2. Cardiac effects, e.g., atrial fibrillation
3. Skeletal effects, e.g., osteoporosis/osteopenia

No consensus exists regarding the management of subclinical hyperthyroidism, but treatment should be individualized.

Treatment is guided by risk of complications such as osteoporosis (especially post-menopausal women) and atrial fibrillation (especially in elderly > 60 years of age).

Patients with subclinical hyperthyroidism due to multi-nodular goitre or functioning adenoma are unlikely to normalize and hence are more likely to benefit from treatment.

### **Detection and Treatment of Complications**

1. Complications from untreated hyperthyroidism:  
Thyroid storm and cardiac complications will need referral to Emergency Department.  
Treatment of hyperthyroidism would address neurologic complications like tremors and myopathy.  
Severe Graves' ophthalmopathy needs referral to Ophthalmologist.  
Thyrotoxic periodic paralysis can cause hypokalaemia which needs preventive beta-blocker.
2. Complications from medical treatment of hyperthyroidism: adverse reaction to anti-thyroid drugs (as above)

### **Urgent and Emergency Situations**

1. Thyroid storm (with rapid heartbeat, high fever, confused and agitated, diarrhoea, possible loss of consciousness)
2. Cardiac: atrial fibrillation, heart failure, angina pectoris
3. Neurologic: thyrotoxic periodic paralysis
4. Agranulocytosis from adverse reaction to anti-thyroid drugs

These require referral to Emergency Department.

## **REFERRALS**

1. Consider referral to Endocrine SOC if adverse reaction to medication, difficulty to achieve euthyroid status despite prolonged treatment or multiple relapses on medication.
2. Refer to Endocrine SOC or pregnancy clinic if pregnancy at any trimester and within first 6 months.
3. Refer to Cardiology if there is cardiac compromise.
4. Refer to Ophthalmology for thyroid eye disease.
5. Atypical presentations, e.g., pain, 6 weeks to 6 months postpartum (thyroiditis).
6. Patients who are on drugs that can affect thyroid function.
  - Prescription drugs e.g., Amiodarone, Lithium
  - Non-prescription drugs or traditional medicines e.g., spirulina, seaweed (iodine-containing) preparations.  
Condition is reversible if these drugs are stopped.
7. Thyroid gland:
  - Solitary nodule
  - Multinodular goitre
  - Tender / inflamed goitre
  - Suspected malignancy

## **ABBREVIATIONS**

|          |                                                 |
|----------|-------------------------------------------------|
| anti-TG  | Thyroglobulin antibodies                        |
| anti-TPO | Thyroid peroxidase antibodies                   |
| fT4      | free T4                                         |
| TFT      | Thyroid function test                           |
| TRAb     | Thyroid Stimulating Hormone Receptor Antibodies |
| TPO Ab   | Thyroid Peroxidase antibodies                   |
| TSH      | Thyroid-Stimulating Hormone                     |
| RAI      | Radioactive iodine                              |

## **REFERENCES**

1. ATA/ AACE Clinical practice guidelines for hypothyroidism in adults. Endocrine Practice. Nov/Dec 2012; Vol. 18 No. 6.
2. Ross, *et al.* 2016 American Thyroid Association guidelines for diagnosis and management of hyperthyroidism and other causes of thyrotoxicosis. Thyroid Oct 2016, 26(10):1343-1421.
3. Jacqueline Jonklass, *et al.* ATA Guidelines for the Treatment of Hypothyroidism. Thyroid Vol 24, No. 12, 2014.
4. British Columbia Medical Association Guidelines & Protocols Advisory Committee on Thyroid disease – Thyroid Function Tests in the Diagnosis and Monitoring of Adults; 1 October 2004.
5. Toft AD. Subclinical hyperthyroidism. New England Journal of Medicine 2001; 345(7):512-516.
6. Orna Diav-Citrin & Asher Ornoy. Teratogen Update: Antithyroid Drugs – Methimazole, Carbimazole, and Propylthiouracil. Teratology 65:38-44 (2002).
7. Bahn RS, Burch HS, Cooper DS, Garber JR, Greenlee CM, Klein IL, Laurberg P, McDougall IR, Rivkees SA, Ross D, Sosa JA, Stan MN 2009. The role of propylthiouracil in the management of Graves' disease in adults: report of a meeting jointly sponsored by the American Thyroid Association and the Food and Drug Administration. Thyroid 19:673–674.d
8. Thyroid Disease: Assessment and Management. NICE Guidelines (NG 145) 20 Nov 2019.
9. Up to Date: accessed in August 2023